

Every pair fails. The three-way form succeeds.

An exact mixed-support ES certificate

The original shell is

$$p = 2671180768668258904496300170662649,$$

$$a = 2^5 431^3 457^3 7229^4 = (p + 135)/4.$$

Retain all 9702 original divisor/channel states $u \mid a^2$ and set

$$D = \frac{135}{\gcd(135, g_c)}, \quad g_E = 4u + 1, \quad g_M = u + a, \quad \beta_{457} = v_{457}(u) - 3.$$

The three observations are

$$x_\alpha = D_\alpha^{-1}, \quad f = (1, x\mathbf{1}_E, x\mathbf{1}_{\beta_{457}>0}),$$

$$J_{ij} = \sum_{\alpha} (x_\alpha - 1/3) f_i(\alpha) f_j(\alpha).$$

Every two-coordinate restriction of J is negative definite. The full form has one positive direction, with exact trial

$$F = \frac{-1 + 36x\mathbf{1}_E + 36x\mathbf{1}_{\beta_{457}>0}}{71}.$$

$$\frac{3}{2} \sum_{\alpha} (x_\alpha - 1/3) F_\alpha^2 = \frac{607462}{1890375} > 0.$$

Failed states contribute nonpositively. At a successful state $|F| \leq 1$. Therefore an original integral ES witness exists.

What makes the difference? The actual overlap $E \cap \{\beta_{457} > 0\}$ supplies the mixed cross entry. Deleting it changes the same trial value to

$$-\frac{1639754}{17013375} < 0.$$

All weights, exponent multiplicities and ordered denominator maps remain. The three components are specified observations of one source, not three independent collections of possible witnesses.

The short preprint and full proof include the exact matrix, actual witnesses, primality certificate, standard-library verifier, and the uniform mixed-filter extension. This is a strict sufficient-test improvement in a known hard shell, not a proof of universal Erdős–Straus occupancy or a historical-priority claim.